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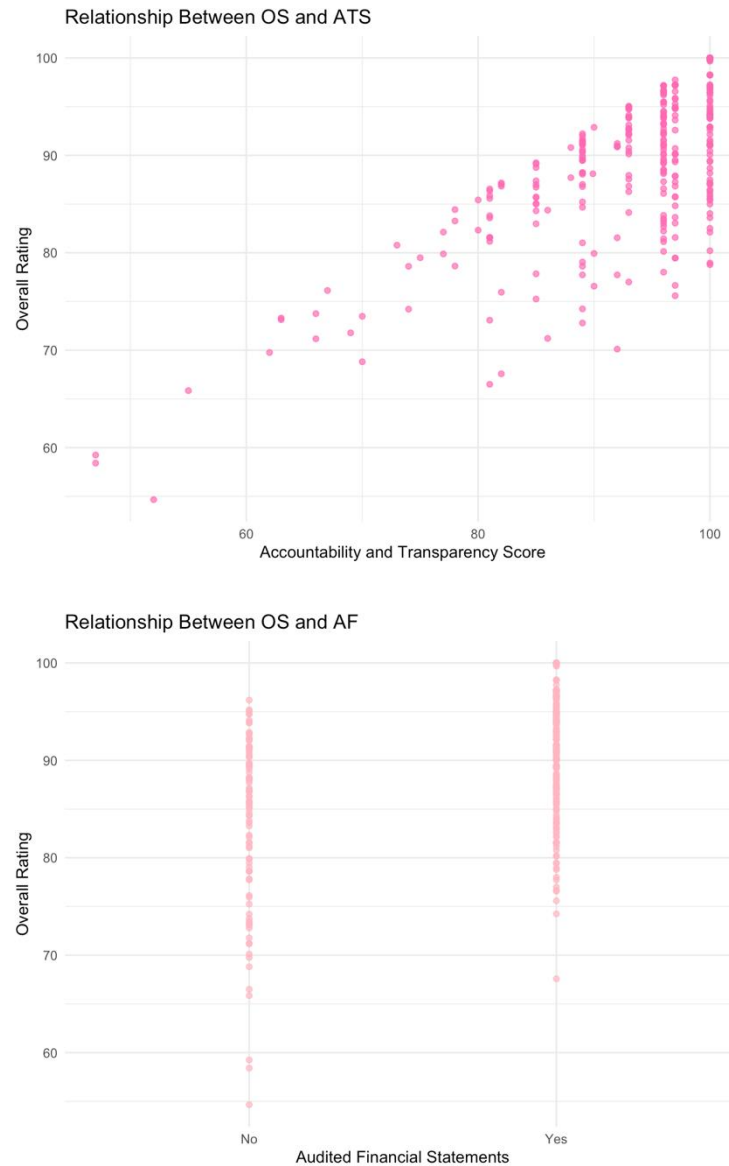
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This report is aimed at analyzing the relationship between a charity's overall rating and, first, its accountability and transparency score and, second, whether audited financial statements are provided on the charity's website. By using the data from charitynavigator.org, we address two research questions: (1) how overall rating is related to accountability and transparency score, and (2) how overall rating is related to whether audited financial statements are provided on the charity's website. This report begins by analyzing the graphical representations of two scatter plots, then combines this with the correlation report generated by R to draw conclusions. Finally, the two results are compared to determine which variable had a stronger relationship with the overall rating.

To visualize the data, we draw two scatterplots by using R. The first plot illustrates the relationship between the overall rating and accountability and transparency score, with the overall rating shown on the y-axis and the accountability and transparency score shown on the x-axis. For the binary variable indicating whether audited financial statements are provided, we do not want it treated as a scaled variable. Therefore, we converted 0 to "No" and 1 to "Yes" in the scatter plot. The resulting plots are shown below:



Observing the first plot, we can conclude that the correlation analysis is appropriate for testing the relationship between overall rating and accountability and transparency score. We can find that a simple curve does not give a better fit for the data than a line does, so a linear model is appropriate. It appears that the overall rating follows a positive linear relationship without obvious outliers. Informally, for the charities with relatively high accountability and transparency scores, their overall rating is also relatively high. The relationship appears to be strong.

For the second plot, a correlation analysis seems also appropriate for examining the relationship between overall rating and whether audited financial statements are provided. Since the variable of audited financial statements is binary, we cannot directly apply linearity to decide whether using correlation analysis is appropriate or not. Instead, we can determine that by observing whether there are notable differences in overall ratings between the yes and no groups and whether there is any extreme outlier. We can conclude for the “no” group, the rating is clustered around the range 70 to 95, while the rating of “yes” group is around the range 80 to 100. Also, there is almost no extreme outlier for both groups. Informally, charities that provide their audited financial statements tend to have relatively higher overall ratings. The relationship seems to be moderate.

By using R to calculate the correlation test for overall rating and accountability and transparency score, we have that $p\text{-value} < 2.2e-16$, which is smaller than 0.05, indicating that the relationship is statistically significant. The value of the correlation between the overall rating and the accountability and transparency score is $r=0.7070044$. According to Cohen’s scale, this value indicates that there is a strong, positive correlation between these two variables, suggesting that charities where the average accountability and transparency score is relatively high tend also to be charities where the overall rating is relatively high.

For the test for the relationship between overall rating and whether audited financial statements are provided, the p-value is less than $3.546e-16$, which is also smaller than 0.05, indicating that the relationship is statistically significant. The value of correlation is $r=0.428312$, which suggests that charities where the audited financial statement is provided tend to have higher overall ratings than those that do not provide audited financial statements. According to Cohen’s scale, the correlation is moderate.

To conclude, both variables are somehow related with a charity's overall rating. The charity that has a higher accountability and transparency score tends to have a higher overall rating. And the charity that provided its audited financial statement also tends to have a higher overall rating. However, the accountability and transparency score shows a stronger relationship with the overall rating than the presence of an audited financial statement, as inferred by its bigger correlation value. This suggests that the difference in overall rating is more closely related to the difference in accountability and transparency score. Also, another key difference lies in the fact that the accountability and transparency score is a scaled variable. Therefore, we can describe the linear relationship between them as either positive or negative. In contrast, the presence of audited financial statements is a binary variable. We can only compare the difference in overall rating between having audited financial statement and not having it without describing a positive or negative linear relationship.